Successive Convexification for Powered Descent Guidance with Time-Varying Mass Properties

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This paper extends the 6 degree-of-freedom successive convexification (SCvx) algorithm discussed in literature to account for time-varying mass properties in the representation of the attitude dynamics. Simulation results of a notional large-size lander performing the last minutes of Lunar powered descent with significant variations in the center-of-mass and moments of inertia over the duration of the trajectory are shown. Differences in the final optimal trajectories generated via SCvx are presented between the constant vs. variable mass property formulations for an in-plane and out-of-plane example.

I. Introduction

The seminal papers by Szmuk et.al in 2017[1] and 2018[2] on the application of successive convexification (SCvx) to the 6-DoF rocket powered landing problem with fixed and free final times introduced a new paradigm in the field of real-time trajectory optimization. These papers leverage the power of convex optimization in generating optimal trajectories that are dynamically feasible while satisfying state and control constraints. Follow-on papers[3–5] extend the SCvx algorithm to use dual quaternions and modified Rodrigues parameters for the attitude representation and account for state-triggered constraints. References [6,7] apply SCvx to the rendezvous and docking problem and landing on Titan. More applications of SCvx are found in Ref. [8].

To the best of the authors' knowledge, the underlying vehicle attitude dynamics used in the SCvx procedure in all the aforementioned references assume constant center-of-mass and moments of inertia and are described by Eqs. (3b) and (5) of Ref. [2]. In most powered descent applications this may be a good approximation. However, vehicle designs such as SpaceX' Lunar Starship[9] have significant variations in the center-of-mass and inertia throughout the trajectory that need to be taken into account in the formulation of the optimal trajectory. The entire Apollo powered descent phase took nearly 12 minutes.[10] During such a long maneuver, large deviations in the vehicle mass properties are expected between the start and end of powered descent for a massive vehicle such as the Lunar Starship. This shortcoming is partially addressed by Refs. [14] and [11] where the inertia is assumed to be an affine function of mass and the center of mass is assumed to be fixed. However, this approximation is crude and lacks mathematical rigor in the formulation of the time-varying attitude dynamics. This paper seeks to fill that gap by leveraging the work performed in Ref. [12] to modify the attitude dynamics portion of Problem 1 of Ref. [2] to account for time-varying mass properties.

The remainder of the paper consists of three sections. Sec.. II presents a summary of the full attitude dynamics, similar to the material presented in Ref. [12]. Sec. III compares time-optimal trajectories for a notional large-sized lander performing the last minutes of Lunar powered descent using fixed and time-varying mass properties (MP) within the SCvx algorithm. Sec. IV provides concluding remarks.

II. Dynamics

The derivation of the complete attitude dynamics accounting for time-varying inertia and moving mass center closely follows Ref. [12]. The dynamic system of interest is modeled as a single rigid body, B, combined

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with n number of bodies $C_1, \ldots, C_n$, each of whose mass can vary with time but is otherwise regarded as rigid; each such body has a single translational degree of freedom relative to B. The empty dry mass of the vehicle is denoted by m_B. C_i is a cylindrical propellant volume of mass m_i that can vary with time. Sloshing is ignored which provides an opportunity for research. The liquid is assumed to occupy a cylindrical volume whose height changes as the propellant is expelled. Approximation of the total mass depletion dynamics is described by Eq. (1) of Ref. [2]. For a typical liquid-fueled rocket consisting of an oxidizer and fuel tank with a mixture ratio, κ, the respective mass flow rate, $\dot{m}$, for the tanks can represented as: $\dot{m}_{oxidizer} = \kappa\dot{m}$ and $\dot{m}_{fuel} = (1-\kappa)\dot{m}$.

The angular momentum in an inertial reference frame, N, of the system S with respect to the system mass center $S^\star$ is denoted by ${}^N\mathbf{H}^{S/S^\star}$ and can be referred to simply as the central angular momentum of S in N. The angular momentum principle, expressed as

$$\frac{{}^N d\,{}^N\mathbf{H}^{S/S^\star}}{dt} = \mathbf{M} \tag{1}$$

states that the time derivative in N, denoted by ${}^N d/dt$, of ${}^N\mathbf{H}^{S/S^\star}$ is equal to the resultant moment $\mathbf{M}$ about $S^\star$ of the external forces applied to S.

The central angular momentum of S in N can be expressed as

$${}^N\mathbf{H}^{S/S^\star} = \underline{\mathbf{I}}^{B/B^\star} \cdot {}^N\boldsymbol{\omega}^B + m_B\,\mathbf{r}^{S^\star B^\star} \times {}^N\mathbf{v}^{B^\star} + \sum_{i=1}^{n}\left(\underline{\mathbf{I}}^{C_i/C_i^\star} \cdot {}^N\boldsymbol{\omega}^{C_i} + m_i\,\mathbf{r}^{S^\star C_i^\star} \times {}^N\mathbf{v}^{C_i^\star}\right) \tag{2}$$

where ${}^N\boldsymbol{\omega}^B$ is the angular velocity of B in N, $\mathbf{r}^{S^\star B^\star}$ is the position vector from $S^\star$ to $B^\star$ (the mass center of B), ${}^N\mathbf{v}^{B^\star}$ is the velocity of $B^\star$ in N, and $\underline{\mathbf{I}}^{B/B^\star}$ is the inertia dyadic of B relative to $B^\star$, or simply the central inertia dyadic of B. The terms in the sum are defined similarly. The velocity ${}^N\mathbf{v}^{C_i^\star}$ of $C_i^\star$ in N can be expressed as

$${}^N\mathbf{v}^{C_i^\star} = {}^N\mathbf{v}^{B^\star} + {}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star C_i^\star} + {}^B\mathbf{v}^{C_i^\star} \qquad (i = 1, \ldots, n) \tag{3}$$

where $\mathbf{r}^{B^\star C_i^\star}$ is the position vector from $B^\star$ to $C_i^\star$ and the velocity ${}^B\mathbf{v}^{C_i^\star}$ of $C_i^\star$ in B is the time derivative in B of the position vector from a point fixed in B to $C_i^\star$ $(i = 1, \ldots, n)$. The velocity terms on the right-hand side of Eq. (2) can be collected and expanded to obtain

$$\begin{aligned}
&m_B\,\mathbf{r}^{S^\star B^\star} \times {}^N\mathbf{v}^{B^\star} + \sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star} \times {}^N\mathbf{v}^{C_i^\star} \\
&\quad = \left(m_B\,\mathbf{r}^{S^\star B^\star} + \sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star}\right) \times {}^N\mathbf{v}^{B^\star} + \sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star C_i^\star} + {}^B\mathbf{v}^{C_i^\star}\right) \\
&\quad = \sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star C_i^\star} + {}^B\mathbf{v}^{C_i^\star}\right)
\end{aligned} \tag{4}$$

where the term in parentheses that precedes ${}^N\mathbf{v}^{B^\star}$ vanishes by virtue of the definition of the system mass center. The following term term in Eq. (4) can be expanded as

$$\sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star C_i^\star}\right) = \sum_{i=1}^{n} m_i\,\mathbf{r}^{S^\star C_i^\star} \times \left[{}^N\boldsymbol{\omega}^B \times \left(\mathbf{r}^{B^\star S^\star} + \mathbf{r}^{S^\star C_i^\star}\right)\right] \tag{5}$$

Now, in view of Eqs. (3.5.31) and (3.6.9) of Ref. [13],

$$\begin{aligned}
m_i\,\mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{S^\star C_i^\star}\right) &= m_i\left[\underline{\mathbf{U}}\left(\mathbf{r}^{S^\star C_i^\star} \cdot \mathbf{r}^{S^\star C_i^\star}\right) - \mathbf{r}^{S^\star C_i^\star}\mathbf{r}^{S^\star C_i^\star}\right] \cdot {}^N\boldsymbol{\omega}^B \\
&\triangleq \underline{\mathbf{I}}^{C_i^\star/S^\star} \cdot {}^N\boldsymbol{\omega}^B \qquad (i = 1, \ldots, n)
\end{aligned} \tag{6}$$

where $\underline{\mathbf{U}}$ is the unit dyadic. $\underline{\mathbf{I}}^{C_i^\star/S^\star}$ is the inertia dyadic relative to $S^\star$ of a (fictitious) particle of mass m_i placed at $C_i^\star$ (Ref. [13], p. 70); in other words, $\underline{\mathbf{I}}^{C_i^\star/S^\star}$ is a parallel axis term that, when added to $\underline{\mathbf{I}}^{C_i/C_i^\star}$, yields the inertia dyadic $\underline{\mathbf{I}}^{C_i/S^\star}$ of C_i relative to $S^\star$.

$$\underline{\mathbf{I}}^{C_i/S^\star} = \underline{\mathbf{I}}^{C_i/C_i^\star} + \underline{\mathbf{I}}^{C_i^\star/S^\star} \qquad (i = 1, \ldots, n) \tag{7}$$

The term on the right-hand side of Eq. (5) that has not yet been dealt with can be developed by appealing once more to the definition of the system mass center.

$$\begin{aligned}\sum_{i=1}^{n} m_i \, \mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star S^\star} \right) &= -m_B \mathbf{r}^{S^\star B^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{B^\star S^\star} \right) \\ &= m_B \mathbf{r}^{S^\star B^\star} \times \left({}^N\boldsymbol{\omega}^B \times \mathbf{r}^{S^\star B^\star} \right) \\ &= m_B \left[\underline{\mathbf{U}} \left(\mathbf{r}^{S^\star B^\star} \cdot \mathbf{r}^{S^\star B^\star} \right) - \mathbf{r}^{S^\star B^\star} \mathbf{r}^{S^\star B^\star} \right] \cdot {}^N\boldsymbol{\omega}^B \\ &\triangleq \underline{\mathbf{I}}^{B^\star/S^\star} \cdot {}^N\boldsymbol{\omega}^B \end{aligned} \tag{8}$$

The inertia dyadic of S relative to $S^\star$ (the central inertia dyadic of S) can be expressed as

$$\underline{\mathbf{I}}^{S/S^\star} = \underline{\mathbf{I}}^{B/B^\star} + \underline{\mathbf{I}}^{B^\star/S^\star} + \sum_{i=1}^{n} \left(\underline{\mathbf{I}}^{C_i/C_i^\star} + \underline{\mathbf{I}}^{C_i^\star/S^\star} \right) \tag{9}$$

Because body C_i can only translate relative to B, ${}^N\boldsymbol{\omega}^{C_i} = {}^N\boldsymbol{\omega}^B$ $(i = 1, \ldots, n)$. In that case, substitution from Eqs. (4), (5), (6), (8), and (9) into Eq. (2) yields

$${}^N\mathbf{H}^{S/S^\star} = \underline{\mathbf{I}}^{S/S^\star} \cdot {}^N\boldsymbol{\omega}^B + \sum_{i=1}^{n} m_i \, \mathbf{r}^{S^\star C_i^\star} \times {}^B\mathbf{v}^{C_i^\star} \tag{10}$$

Applying the angular momentum principle yields the complete attitude dynamics:

$$\begin{aligned} &\frac{{}^B d\underline{\mathbf{I}}^{S/S^\star}}{dt} \cdot {}^N\boldsymbol{\omega}^B + \underline{\mathbf{I}}^{S/S^\star} \cdot \frac{{}^B d\, {}^N\boldsymbol{\omega}^B}{dt} + {}^N\boldsymbol{\omega}^B \times \left(\underline{\mathbf{I}}^{S/S^\star} \cdot {}^N\boldsymbol{\omega}^B \right) \\ &+ \sum_{i=1}^{n} \left[\dot{m}_i \, \mathbf{r}^{S^\star C_i^\star} \times {}^B\mathbf{v}^{C_i^\star} + m_i \left(\frac{{}^B d\mathbf{r}^{S^\star C_i^\star}}{dt} + {}^N\boldsymbol{\omega}^B \times \mathbf{r}^{S^\star C_i^\star} \right) \times {}^B\mathbf{v}^{C_i^\star} \right. \\ &\left. + m_i \, \mathbf{r}^{S^\star C_i^\star} \times {}^B\mathbf{a}^{C_i^\star} + m_i \, \mathbf{r}^{S^\star C_i^\star} \times \left({}^N\boldsymbol{\omega}^B \times {}^B\mathbf{v}^{C_i^\star} \right) \right] = \mathbf{M} \end{aligned} \tag{11}$$

where the acceleration ${}^B\mathbf{a}^{C_i^\star}$ of $C_i^\star$ in B is the time derivative in B of ${}^B\mathbf{v}^{C_i^\star}$ and $\dot{m}_i$ is the mass flow rate of the ith tank. The term, ${}^B\mathbf{a}^{C_i^\star}$, can be related to the engine throttle rate. The contributions of the moving propellant to the attitude dynamics are magnified if the fuel mass fraction and mass flow rates are both significant. It is worth noting that terms in the sum vanish when certain vectors are parallel to each other, such as $\mathbf{r}^{S^\star C_i^\star}$ and ${}^B\mathbf{v}^{C_i^\star}$ in the case of a tandem concentric tank configuration like most conventional rockets. The time derivative in B of $\underline{\mathbf{I}}^{S/S^\star}$ in Eq. (9) can be expressed as

$$\frac{{}^B d\underline{\mathbf{I}}^{S/S^\star}}{dt} = \underline{\mathbf{0}} + \frac{{}^B d\underline{\mathbf{I}}^{B^\star/S^\star}}{dt} + \sum_{i=1}^{n} \left(\frac{{}^B d\underline{\mathbf{I}}^{C_i/C_i^\star}}{dt} + \frac{{}^B d\underline{\mathbf{I}}^{C_i^\star/S^\star}}{dt} \right) \tag{12}$$

where $\underline{\mathbf{0}}$ is the zero dyadic. In the formulation of the trajectory optimization problem with time-varying MP, Eq. (11) would replace the standard Euler's equation with everything else identical to Problem 1 of Ref. [2].

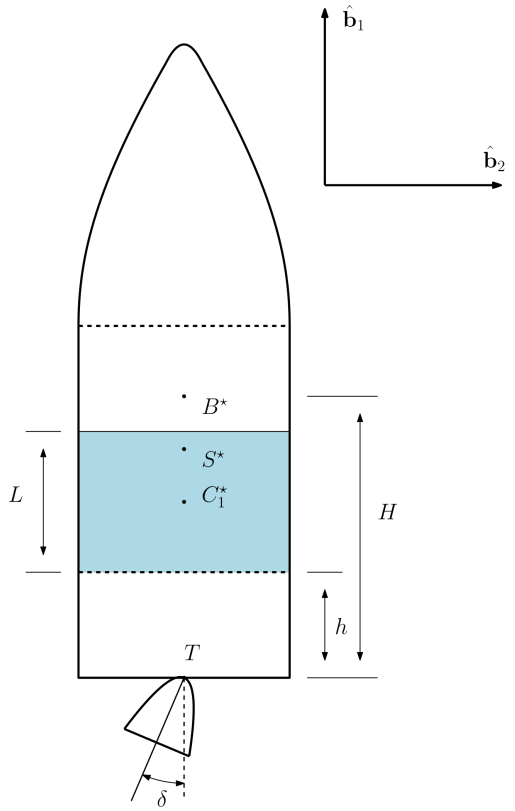


Figure 1: Nomenclature

A schematic of a notional rocket with a single propellant tank is shown in Figure 1 with the origin of the coordinate system fixed at $B^\star$. H is the distance from $B^\star$ to the thrust application point, T, L is the height of the propellant inside the tank, and h represent the offset between T and the aft end of the tank. Position vectors from $B^\star$ to ${C_1}^\star$, $B^\star$ to $S^\star$, $S^\star$ to ${C_1}^\star$, and $S^\star$ to T (thrust moment arm) are presented in Eqs. (13) through (16).

$$\mathbf{r}^{B^\star C_1^\star} = \left(-H + h + \frac{m_1}{2S^2}\right)\hat{\mathbf{b}}_1 \tag{13}$$

$$\mathbf{r}^{B^\star S^\star} = \frac{m_1}{m_T}\left(-H + h + \frac{m_1}{2S^2}\right)\hat{\mathbf{b}}_1 \triangleq \lambda\hat{\mathbf{b}}_1 \tag{14}$$

$$\mathbf{r}^{S^\star C_1^\star} = \left(-H + h + \frac{m_1}{2S^2} - \lambda\right)\hat{\mathbf{b}}_1 \tag{15}$$

$$\mathbf{r}^{S^\star T} = (-H - \lambda)\,\hat{\mathbf{b}}_1 \tag{16}$$

where $S^2 = \rho\pi R^2$, ρ is the density of the liquid, R is the radius of the tank, and $m_T = m_B + \sum_{i=1}^{n} m_i$. Terms that make up $\underline{\mathbf{I}}^{S/S^\star}$ are shown in Eqs. (17) to (19).

$$\underline{\mathbf{I}}^{B^\star/S^\star} = m_B \begin{bmatrix} 0 & 0 & 0 \\ 0 & \lambda^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix} \tag{17}$$

$$\underline{\mathbf{I}}^{C_1/C_1^\star} = m_1 \begin{bmatrix} \frac{1}{2}R^2 & 0 & 0 \\ 0 & \frac{1}{12}(3R^2 + m_1^2/S^4) & 0 \\ 0 & 0 & \frac{1}{12}(3R^2 + m_1^2/S^4) \end{bmatrix} \tag{18}$$

$$\underline{\mathbf{I}}^{C_1^\star/S^\star} = m_1 \begin{bmatrix} 0 & 0 & 0 \\ 0 & \left[-H + h + m_1/(2S^2) - \lambda\right]^2 & 0 \\ 0 & 0 & \left[-H + h + m_1/(2S^2) - \lambda\right]^2 \end{bmatrix} \tag{19}$$

III. Results

Successive convexification of a notional large-size lander performing the approach phase of Lunar powered descent using fixed and time-varying MP formulations of the attitude dynamics are presented. For the time-varying MP formulation, Eq. (11) replaces the standard Euler's equation used in the algorithm from [2]. The approach phase is generally a more dynamic portion of the flight profile with more aggressive attitude changes compared to the braking phase. The approach phase also tends to have more stringent requirements on the vehicle states such as line-of-sight, maximum tilt angle, and maximum angular velocity requirements. For simplicity, the lander is assumed to have a single oxidizer tank ($\kappa = 1$). Algorithm parameters, described in [2], are presented in Table 1 along with the initial and terminal conditions. Parameters describing the properties of the variable MP vehicle are presented in Table 2. Average values of the inertia and moment arm from the variable MP case are used as the values in the constant MP case.

Table 1: Algorithm Parameters and Boundary Conditions

Algorithm Parameters		Boundary Conditions	
Parameter	Value	Parameter	Value
ω_σ	4×10^{-2}	$m_i(t_0)$	$4.25{\times}10^6$ kg
ω_ν	10^5	$r(t_o)$	[4000 6000 0] m
ω_Δ	10^{-3}	$v(t_o)$	[-80 -80 0] m/s
$\omega_{\Delta\sigma}$	10^{-1}	$q(t_o)$	[0.976 0 0 0.216]
K	50	$\omega(t_o)$	[0 0 0] deg/s
γ_{gs}	20 deg	$r(t_f)$	[10 0 0] m
θ_{max}	35 deg	$v(t_f)$	[0 0 0] m/s
ω_{max}	5 deg/s	$q(t_f)$	[1 0 0 0]
		$\omega(t_f)$	[0 0 0] deg/s

Table 2: Vehicle Properties

Parameter	Value
m_B	$3.25{\times}10^6$ kg
$[\mathrm{I}]^{B/B^\star}(2,2)$	$4.15{\times}10^7$ kg·m^2
H	-12 m
h	0 m
ρ	490 kg/m^3
I_{sp}	330 s
T_{max}	2250 kN
T_{min}	600 kN
δ_{max}	10 deg

Time histories of thrust moment arm, $\mathbf{r}^{S^\star T}$, and transverse moment of inertia are shown in Figs. 2 and 3 for the variable MP formulation. $\mathbf{r}^{S^\star T}$ starts out at an initial value of -9.5 m and decreases by nearly -0.7 m towards the end of the trajectory as propellant is being consumed. $\underline{\mathbf{I}}^{S/S^\star} \cdot \hat{\mathbf{b}}_3$ decreases from 6.09e7 kg-m^2 to 5.34e7 kg-m^2. In this example, the assumption of a linear variation between mass and transverse inertia seems to be a good approximation. The combined effect of the increasing thrust moment arm and decreasing

transverse moment of inertia suggests that the vehicle becomes more agile over time.

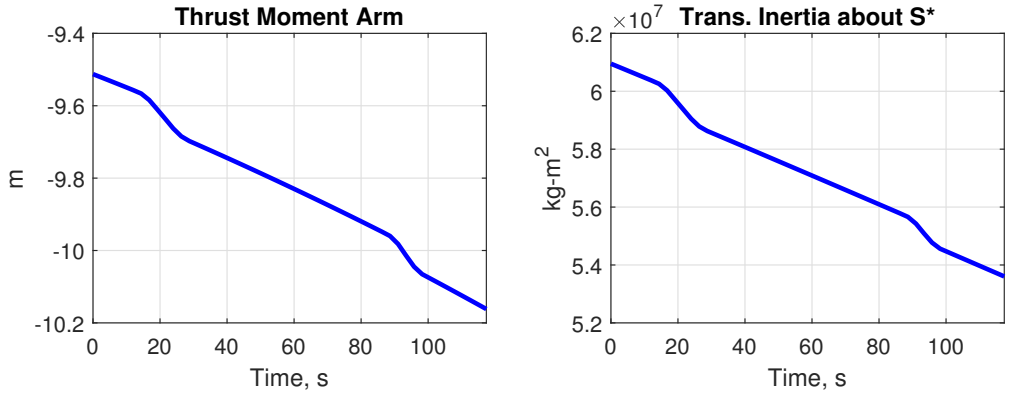


Figure 2: Planar Example: Thrust Moment Arm and Transverse Inertia Variations

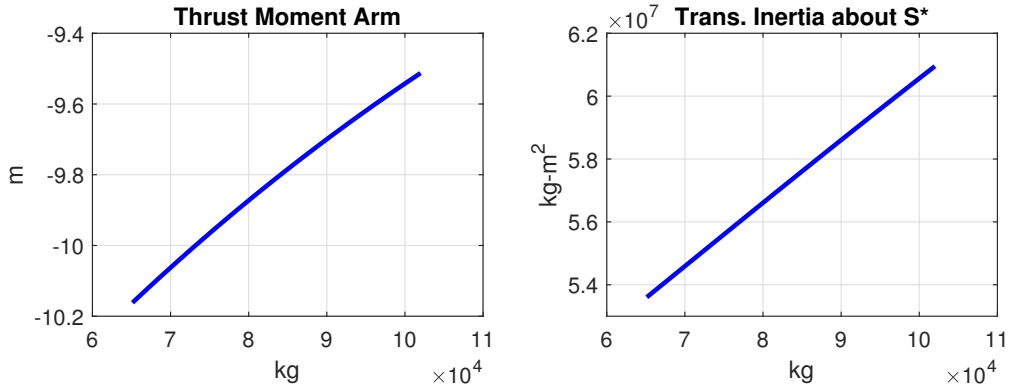


Figure 3: Planar Example: Thrust Moment Arm and Transverse Inertia vs. Mass

For comparison, the fixed MP formulation used the average values of $\mathbf{r}^{S^\star T}$ and $\underline{\mathbf{I}}^{S/S^\star}$. Figures 4 and 5 show comparisons of the time-optimal trajectories between the fixed and time-varying formulations of the attitude dynamics for a planar case. The trajectories are nearly identical, differing by tens of meters at most. The time-of-flight and ΔV are nearly identical, differing by 0.15 s and 0.05 m/s respectively. The ΔV for the constant MP case is 295.5 s. Gimbal angle usage, vehicle tilt, thrust magnitude, and angular velocity norm plots illustrate small differences. Some intuition can be drawn from the angular velocity norm plot as the fixed MP formulation shows slightly higher values at the beginning of the trajectory where the vehicle is more agile compared to the variable MP formulation due to the constant MP case using the average mass properties from the variable MP case.

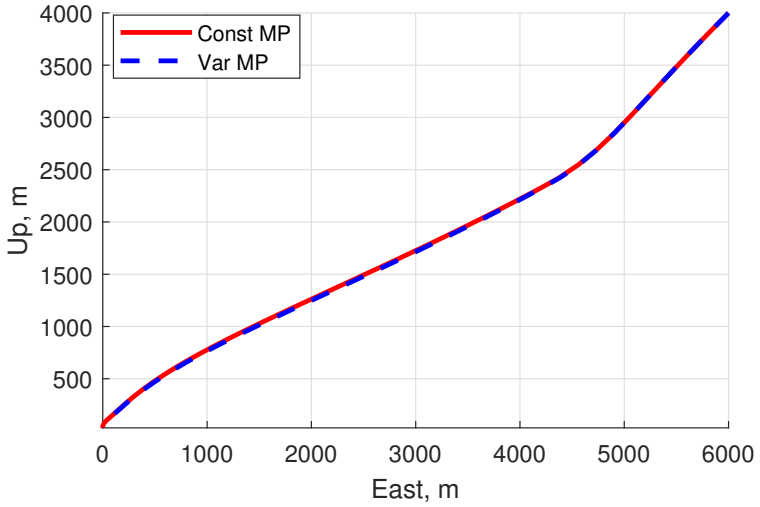


Figure 4: Planar Example: Trajectory Time History, Fixed vs. Variable MP Formulation

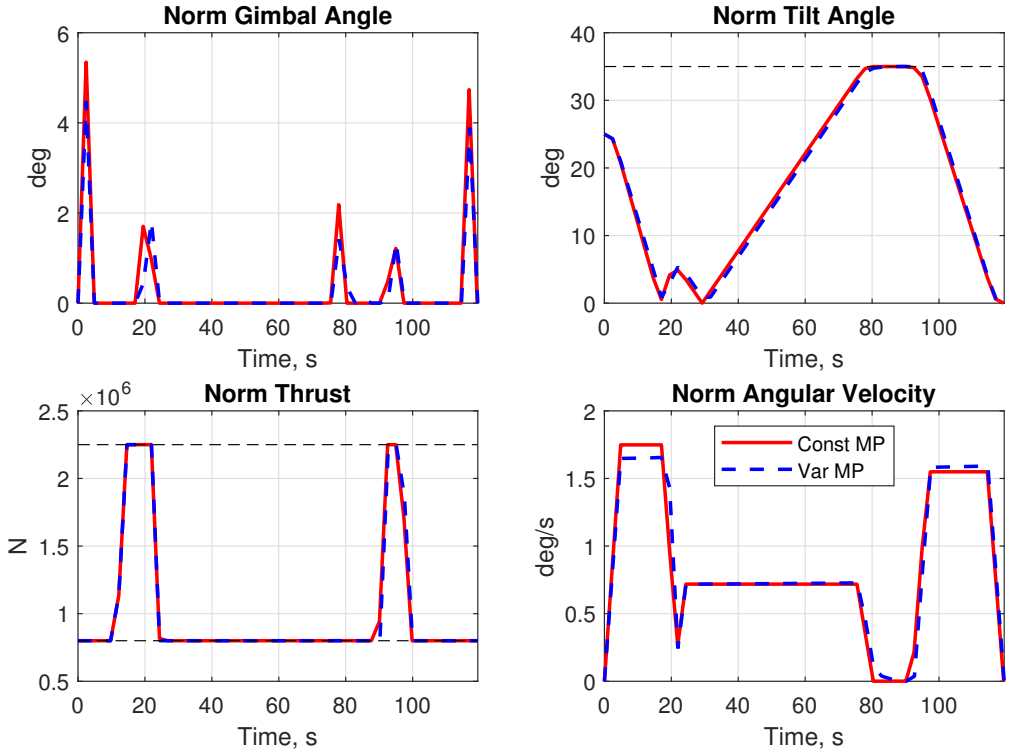


Figure 5: Planar Example: Vehicle Dynamics, Fixed vs. Variable MP Formulation

Figures 6 and 7 show comparisons of the time-optimal trajectories between the fixed and time-varying formulations for an initial crossrange error of 500 m. Differences in the trajectories are larger than the zero crossrange error case with the variable mass formulation exhibiting a slightly shallower glideslope angle after the first 20 sec. The time-of-flight and ΔV differ by slightly more compared to the zero crossrange error case but are also nearly identical. The time-flight differs by 0.5 s where as the ΔV differs by 1 m/s compared to the constant MP case with $\Delta V = 293$ m/s. Compared to the zero crossrange error case, the gimbal angle usage, vehicle tilt, thrust magnitude, and angular velocity norm plots display more noticeable differences.

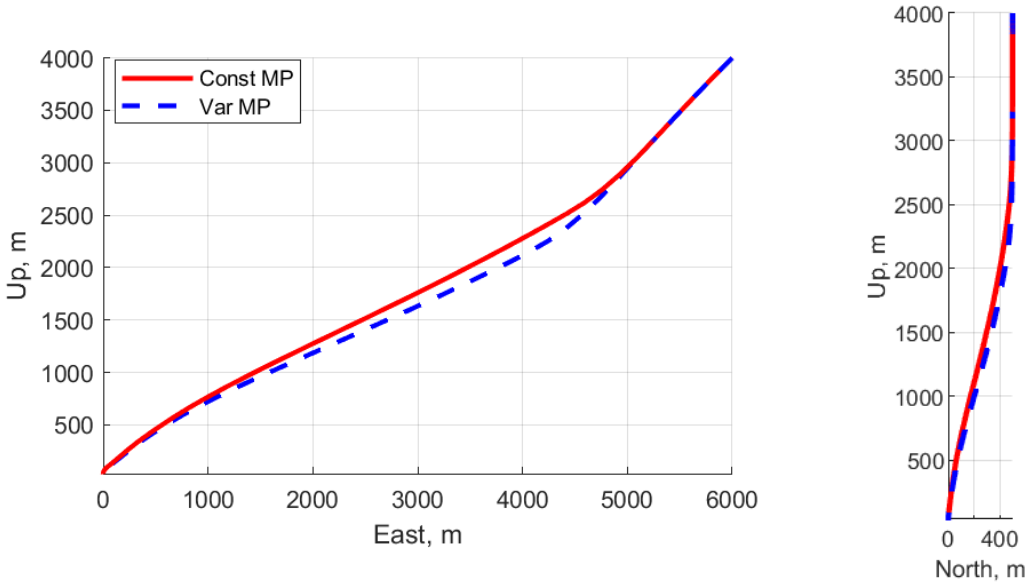


Figure 6: Nonplanar Example: Trajectory Time History, Fixed vs. Variable MP Formulation

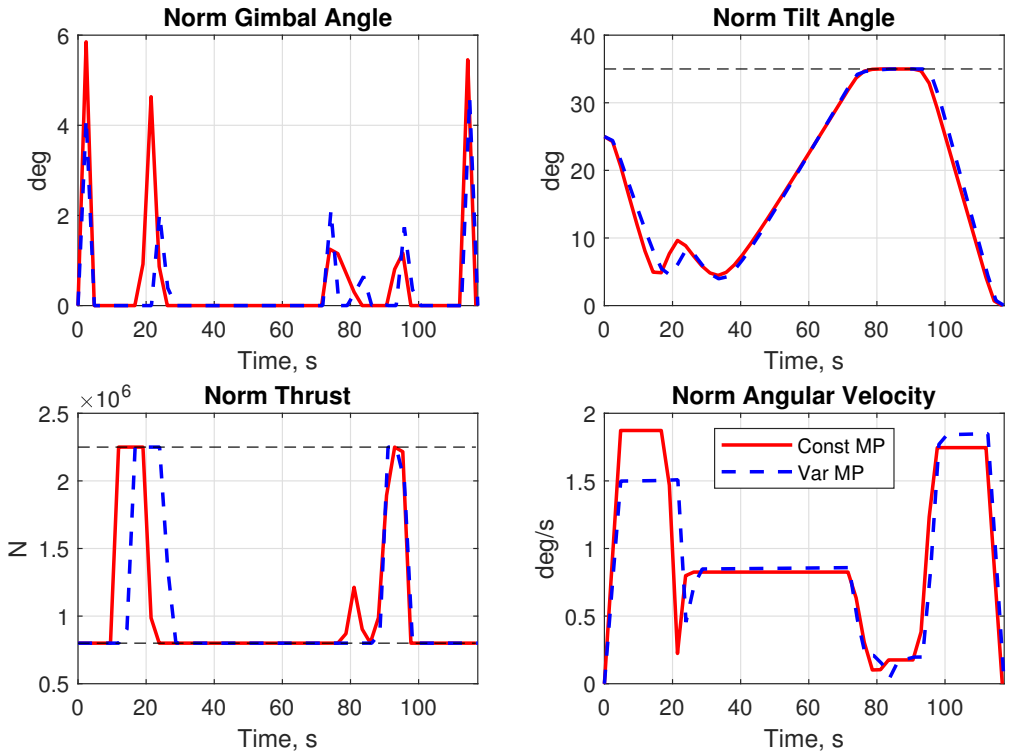


Figure 7: Nonplanar Example: Vehicle Dynamics, Fixed vs. Variable MP Formulation

IV. Conclusions

In this paper, a rigorous derivation of the attitude dynamics that accounts for time-varying mass properties was implemented in the 6 degree-of-freedom successive convexification (SCvx) framework. The example shown in the paper is a large-size lander performing the last 2 minutes of Lunar powered descent. De-

spite appreciable changes in the system mass-center and transverse moment of inertia, the constant mass formulation appeared to produce similar results compared to the variable mass property formulation. The resultant time-of-flight and ΔV values were nearly identical. Both in-plane and out-of-plane scenarios were explored, the variable mass property formulation yielded a slightly lower glideslope angle in the middle of trajectory. The simplified assumption of linear variation between the vehicle mass and transverse moment of inertia appears to be a good approximation for a typical rocket-like configuration with the propellant tanks symmetrically located along the vehicle centerline. However, the methodology described in this paper can be applied to non-conventional vehicle configurations with significant variations in the products-of-inertia or lateral variations in the center-of-mass.

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